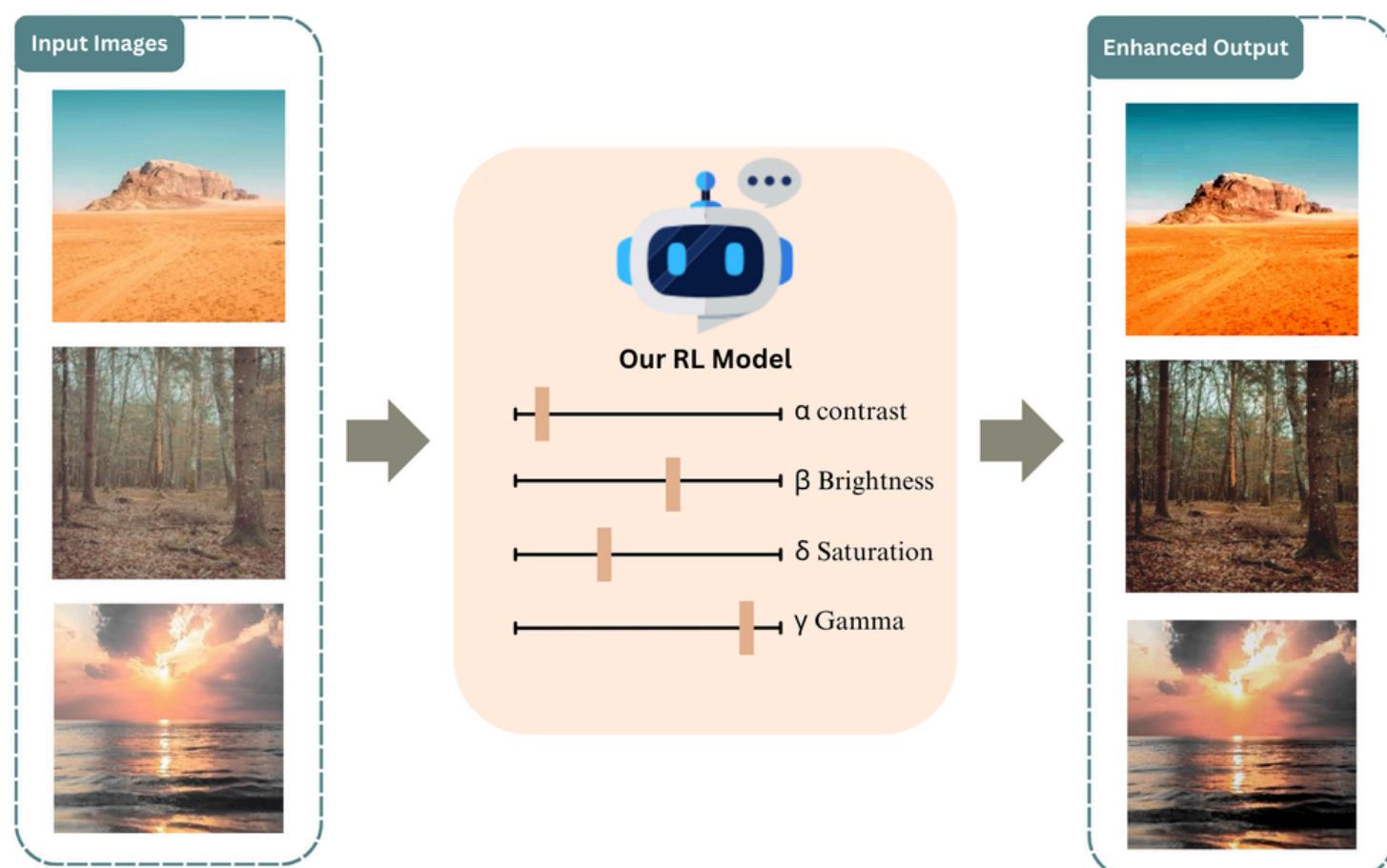




Reinforcement Learning for Scenic Photo Parameter Adjustment with TOPIQ Reward

Hsi-An Chen, Shih-Chen Tseng, Chih-Hsuan Chen

Task



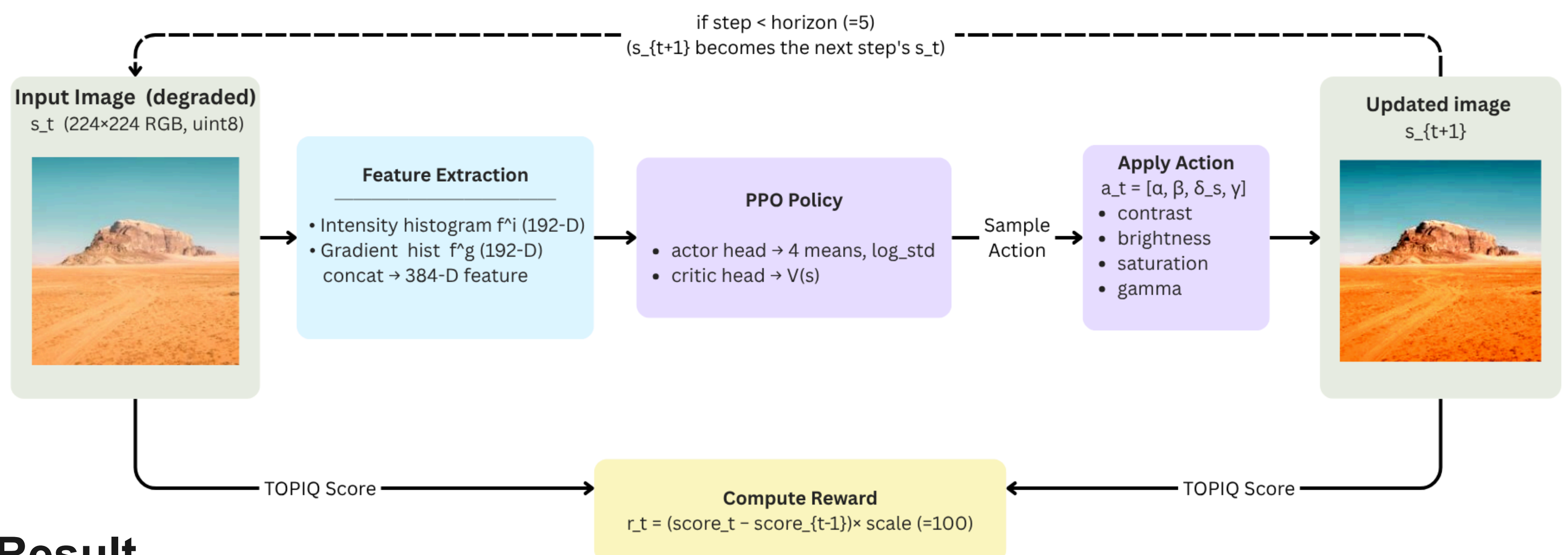
Motivation

Method	Requires GT	Method Type	Input Type
Star Enhancer	Yes	ML	RGB images
DRL-ISP	Yes	RL	RAW images
Ours	No	RL	RGB images

- **No ground truth required.** TOPIQ is a no-reference quality metric, so our policy learns from the image alone, no labeled pairs or reference styles to collect.
- **Reinforcement learning instead of supervised mapping.** PPO lets the agent discover enhancement trajectories that maximize perceived quality, rather than imitating one fixed answer per image.
- **RGB input instead of RAW input.** Most mobile photos are stored as RGB images, while RAW data is often unavailable and cannot be reliably recovered from JPEG.

Method

One episode (horizon = 5 steps)



Result

Method	Input TOPIQ	Output TOPIQ	Improvement
DRL-ISP Baseline	59.5785	45.2433	-14.3352
Star Enhancer Baseline	59.5785	58.2696	-1.3089
Ours (PPO + TOPIQ Reward)	59.5785	61.1498	+1.5713

Example 1: Contrast Degradation



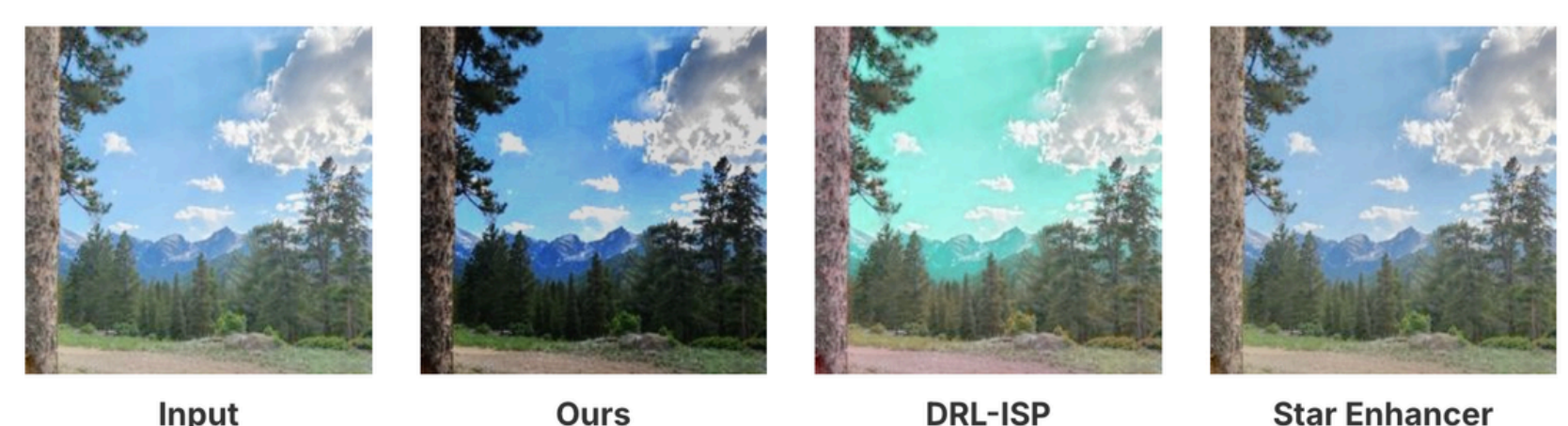
Example 2: Brightness Degradation



Example 3: Saturation Degradation



Example 4: Gamma Degradation



Example 5: All-Parameter Degradation

